MAGIVOX AI

# CSS - A to Z

Selectors | Box Model | Flexbox | Grid | Responsive Design | Animation | Interview

**How to use this guide**
Study one topic, type the example yourself, practise without copying, then take the MAGIVOX topic test. After all modules, take the final course exam.

## Learning outcomes

- CSS Foundation
- Selectors
- Units and Colors
- Box Model
- Typography
- Display and Position

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# 1. CSS Foundation

### What to learn

- Inline/internal/external CSS
- Syntax
- Cascade
- Inheritance
- Specificity

### Practice checkpoint

Explain inline/internal/external css in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

# 2. Selectors

### What to learn

- Element/class/id
- Attribute selectors
- Descendant/child
- Pseudo-classes
- Pseudo-elements

### Practice checkpoint

Explain element/class/id in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

# 3. Units and Colors

### What to learn

- px, %, em, rem, vw, vh
- hex/rgb/hsl
- CSS variables

### Practice checkpoint

Explain px, %, em, rem, vw, vh in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

# 4. Box Model

### What to learn

- Content
- Padding
- Border
- Margin

- box-sizing

### Practice checkpoint

Explain content in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

## 5. Typography

### What to learn

- font-family
- font-size/weight
- line-height
- text alignment
- Readable measure

### Practice checkpoint

Explain font-family in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

## 6. Display and Position

### What to learn

- block/inline/inline-block
- position relative/absolute/fixed/sticky
- z-index

### Practice checkpoint

Explain block/inline/inline-block in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

## 7. Flexbox

### What to learn

- Main/cross axis
- justify-content
- align-items
- gap
- flex-grow/shrink/basis

### Practice checkpoint

Explain main/cross axis in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

## 8. CSS Grid

### What to learn

- Rows/columns
- fr unit
- repeat/minmax
- grid areas
- auto-fit/auto-fill

### Practice checkpoint

Explain rows/columns in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

## 9. Responsive Design

### What to learn

- Mobile-first
- Media queries
- Fluid widths
- Responsive images

### Practice checkpoint

Explain mobile-first in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

## 10. Forms and UI States

### What to learn

- focus/hover/disabled
- Accessible focus rings
- Validation states

### Practice checkpoint

Explain focus/hover/disabled in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

## 11. Transitions and Animation

### What to learn

- transition
- transform
- @keyframes
- prefers-reduced-motion

### Practice checkpoint

Explain transition in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

# 12. Architecture

## What to learn

- Reusable classes
- Component styles
- Naming
- Avoiding specificity wars

## Practice checkpoint

Explain reusable classes in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

# 13. Project

## What to learn

- Responsive dashboard layout
- Cards
- Sidebar
- Mobile navigation
- Dark mode idea

## Practice checkpoint

Explain responsive dashboard layout in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

# 14. Interview Preparation

## What to learn

- Box model
- specificity
- flex vs grid
- position types
- rem vs em
- pseudo-class vs pseudo-element

## Practice checkpoint

Explain box model in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

# Worked Examples

### Flexbox

```
.row { display:flex; gap:1rem; align-items:center; }
```

### Grid

```
.grid { display:grid; grid-template-columns:repeat(auto-fit,minmax(220px,1fr));
gap:1rem; }
```

### Responsive

```
@media (max-width: 768px) { .sidebar { display:none; } }
```

# Interview Answer Pattern

Use this structure for technical questions:

- 1. Definition - say what it is in one or two lines.
- 2. Why - explain why it is useful.
- 3. Example - give a small real or code example.
- 4. Difference/common mistake - compare with a related concept when relevant.
- 5. Be truthful - do not claim tools or experience you have not used.

# Final Course Checklist

- I can explain the foundation without notes.
- I completed examples by typing/doing them myself.
- I completed topic tests and reviewed wrong answers.
- I completed at least one mini project or practical exercise.
- I can answer common interview questions in simple language.
- I am ready for the MAGIVOX 50-question final exam.